

Educations

Ph.D. in Electrical and Electronic Engineering Supervised by Prof. Lihua Xie	Aug 2026 - present Nanyang Technological University (NTU), Singapore
Master Degree in Control Engineering and Automation Supervised by Prof. Tung Lam Nguyen	Jul 2024 - Jul 2026 Hanoi Univ of Science and Technology (HUST), Hanoi, Vietnam
Bachelor Degree in Control Engineering and Automation Supervised by Prof. Tung Lam Nguyen	Oct 2020 - Apr 2024 Hanoi Univ of Science and Technology (HUST), Hanoi, Vietnam

Work experiences

R&D Robotics Engineer at VinRobotics, Vingroup Full-time, supervised by Dr. Minh Nhat Vu	Sep 2025 - Jul 2026 Hanoi, Vietnam
Research Project Assistant at HUST Contract (Hybrid), supervised by PhD. Chinh Hoang Duc and Assoc.Prof.PhD. Tung Lam Nguyen	Oct 2021 - Dec 2026 Hanoi, Vietnam

Selected publications

- [6] Nguyen, T. T. H., **Vu, D. C.**, Pham, M. D., Nguyen, T. L., & Nguyen, T. V. A. (2026). Unifying hierarchical sliding mode control and control barrier function for tilt angle constraint of a ball-balancing robot. *IET Cyber-Systems and Robotics*. [[pdf](#)]
- [5] **Vu, D. C.**[†], Tran, S.[†], Nguyen, T. L., & Hoang, D. C. (2025). Glocal trajectory generation and tracking control for AUVs with optimal coverage sensor networks. *Ocean Engineering*. [[pdf](#)] [†]equal contribution
- [4] **Vu, D. C.**, Nguyen, D. H., Vu, M. N., & Nguyen, T. L. (2025). Lagrangian-based modeling and safety-critical controls for Stewart platforms under marine operations. *Ocean Engineering*, 340, 122142. [[pdf](#)]
- [3] Pham, M. D., **Vu, D. C.**, Nguyen, T. T. H., Nguyen, T. V. A., Vu, M. N., & Nguyen, T. L. (2025). CBFs-based model predictive control for obstacle avoidance with tilt angle limitation for ball-balancing robots. *IEEE Access*. [[pdf](#)]
- [2] **Vu, D. C.**, Nguyen, D. H., & Nguyen, T. L. (2024). A novel approach of consensus-based finite-time distributed sliding mode control for Stewart platform manipulators motion tracking. *Results in Engineering*. [[pdf](#)]
- [1] **Vu, D. C.**, Pham, M. D., Nguyen, T. T. H., Nguyen, T. V. A., & Nguyen, T. L. (2024). Time-optimal trajectory generation and observer-based hierarchical sliding mode control for ballbots with system constraints. *International Journal of Robust and Nonlinear Control*. [[pdf](#)]

Honours & awards

NTU Research Scholarship	Nanyang Technological University (NTU), Singapore
Master, PhD Scholarship Programme	Vingroup Innovation Foundation (VINIF)
Best Thesis Defense Award	Hanoi University of Science and Technology

Academic activities

- Invited review for**
- *Nonlinear Dynamics*
 - *Ocean Engineering*
 - *IEEE Access*
 - *IEEE Robotics and Automation Letters (RA-L)*
 - *International Conference on Intelligent Robots and Systems (IROS 2026)*
 - *IEEE RO-MAN 2026*
 - *Humanoid 2026*